

Task 4:

Prevention of tax evasion through electronic means in Peru

1. Selection of 10 Key Articles via Boolean Search String Executed Search String (Scopus Syntax):

(TITLE-ABS-KEY ("tax evasion" OR "evasión tributaria") AND TITLE-ABS-KEY ("big data" OR "electronic invoice*" OR "facturación electrónica") AND TITLE-ABS-KEY ("it audit*" OR "tax law" OR "derecho tributario") AND TITLE-ABS-KEY ("Peru" OR "SUNAT"))

Final Selected Mapped Corpus (n = 10)

- Anjarwi, A. W. (2026). The digital transformation of tax audits: how AI, big data, blockchain, and advanced analytics are reshaping tax evasion detection. *Journal of Business Analytics*, 1–12. <https://doi.org/10.1080/2573234X.2026.2644363>
- Audrey, Chris (August 11, 2024). Artificial Intelligence in Detecting Tax Fraud. <http://dx.doi.org/10.2139/ssrn.6027334>
- Belahouaoui, R., & Alm, J. (2025). Tax Fraud Detection Using Artificial Intelligence-Based Technologies: Trends and Implications. *Journal of Risk and Financial Management*, 18(9), 502. <https://doi.org/10.3390/jrfm18090502>
- Bhavana, A., Akshatha, Y., Madhusudhan, M. V., Girish, S. C., Impa, B. H., & Pasha, A. (2026). Predictive analysis of income tax fraud using deep learning techniques. *African Journal of Science, Technology, Innovation and Development*, 1–13. <https://doi.org/10.1080/20421338.2026.2696940>
- Chao Lin, Kaili Ma, Xing Zhang (2025). Where technology meets tax: The impact of digital tax administration on tax incentive take-up, *Economics Letters*, Volume 247, 2025, 112118, ISSN 0165-1765. <https://doi.org/10.1016/j.econlet.2024.112118>.
- Gaoya Song, Quan Li (2025). Big data in tax enforcement and trade credit: Evidence from China, *Research in International Business and Finance*, Volume 76, 2025, 102812, ISSN 0275-5319. <https://doi.org/10.1016/j.ribaf.2025.102812>.
- Heinemann, M., Stiller, W. (2025). Digitalization and cross-border tax fraud: evidence from e-invoicing in Italy. *Int Tax Public Finance* 32, 195–237. <https://doi.org/10.1007/s10797-023-09820-x>
- Liu M, Zhang W, Kuang Y (2026), "How big data tax governance reduces excessive deposits and loans? Evidence from Chinese listed firms". *Review of Accounting and Finance*, Vol. 25 No. 3 pp. 344–363. <https://doi.org/10.1108/RAF-09-2025-0360>
- Soto Galarza, I. A., Rivera S. D., Barrera Rea, V. F., & Sandoval Medina, B. A. (2026). Impacto de los algoritmos de inteligencia artificial en la detección de evasión fiscal: Retos de regulación tributaria y garantías procesales tributarias digitales del contribuyente. *Revista Científica Publicar Ciencia*, 8(1). <https://doi.org/10.67169/publicarciencia.v8i1.53>
- Yang L (2025) Predictive modeling of tax compliance risks: A comparative study of machine learning approaches. *PLoS One* 20(9): e0331715. <https://doi.org/10.1371/journal.pone.0331715>

2. PRISMA 2020 Flow Diagram (mathematically consistent)

Fase 1: identification

Database Identification (n = 145) - Scopus: 45 - Web of science: 35 - Scielo/Redalyc: 40 - Alicia CONCYTEC: 25
--

Pre-Screening Deduplication (n = 50)
 - Automated duplicate records removed: 50
 (145 - 50 = 95)

Fase 2: screening

Title & Abstract Screening (n = 95)

Title/Abstract Excluded (n = 65)
 (Traditional physical smuggling/contraband lacking digital or systemic audit tracking)
 (95 - 65 = 30)

Fase 3: eligibility

Full-Text Reports Evaluated (n = 30)

Full-Text Excluded (n = 20)
 Granular justification:
 - Incompatibility with Peruvian Tax Code: 8
 - Pure computer science without audit frameworks: 7
 - Anacronism (published prior to 2015): 5
 (30 - 20 = 10)

Fase 4: inclusion

Studies Included in Meta-Synthesis (n = 10)

Mathematical check: (145 - 50 = 95) screened; (95 - 65 = 30) eligible; (30 - 20 = 10) final included studies.

3. Gap Analysis Table (Categorized & Citation-Backed)

Gap Category	Current State in Literature	Identified Gap / Limitation	Backed by (Citations)
Governance, Regulation & Rights	Tax administrations are aggressively deploying predictive AI models to catch fraud.	Complete absence of binding legal frameworks, oversight architecture, and digital procedural guarantees to protect taxpayer rights.	Soto Galarza et al. (2026); Anjarwi (2026)
Algorithmic Opacity & Bias	High-performing ensemble models (e.g., Random Forest, XGBoost) achieve up to 93% accuracy.	Lack of explicit requirements for "Explainable AI" (XAI) or transparency standards to audit algorithmic bias and "black-box" decision-making.	Soto Galarza et al. (2026); Anjarwi (2026); Yang (2025)
Infrastructure & Interoperability	Implementation of massive centralized data systems (e.g., China's Golden Tax III) or domestic e-invoicing.	High data fragmentation, technical capacity constraints, and lack of secure cross-border data exchange frameworks, especially in developing nations.	Anjarwi (2026); Heinemann & Stiller (2025)
Methodology & Contextual Shift	Heavy reliance on predictive structural indicators (like the Trade Data Gap) and historical correlation data.	Lack of primary empirical validation, strict causal inference models, and cross-regional transferability validation due to heavy regional data constraints.	Yang (2025); Anjarwi (2026)